33rd Tree Labs: The Singularity Protocol

A Byzantine-Resilient Verifiable DePIN Compute Fabric on Solana

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ABSTRACT

As autonomous artificial intelligence agents assume direct fiduciary stewardship over capital, centralized cloud infrastructure introduces fatal vulnerabilities: memory bus snooping, unverified output manipulation, and administrative censorship. While proof-of-work pioneered trustless coordination, evaluating arbitrary hash preimages squanders gigawatts of power on zero computational utility. We present 33rd Tree Labs and the Singularity Protocol, an off-chain verifiable compute fabric settled natively on Solana. The network replaces hash lotteries with Proof of Intelligence (PoI), an unforgeable physical GPU admission gate requiring bit-exact evaluations of standard transformer layers verified via BN254 polynomial commitments, while slot proposals scale with stake among verified nodes. To eliminate hardware vendor single points of failure, confidential execution is governed by a Heterogeneous Multi-TEE Attestation Quorum (Intel TDX, AMD SEV-SNP, and NVIDIA Hopper CC, 2-of-3 threshold) paired with an autonomous Succinct ZK-Fallback cascade. Model weights and activations execute under Calibrated SmoothQuant FP8 (OCP E4M3/E5M2) bit-exact integer reduction trees, guaranteeing bounded error $\Delta\mathcal{L} \leq 1.2 \times 10^{-4}$ on 70B+ parameter Mixture-of-Experts (MoE) topologies. Validator overlays embed into a non-Euclidean Exponential Plunge Hyperboloid manifold $\mathbb{H}^2$ with negative Gaussian curvature that strictly bounds network diameter to ≤ 8 hops under 40% adversarial churn and exponentially raises Sybil Eclipse attack costs. Client inference latency is decoupled through a Three-Tier Finality Pipeline delivering instant speculative execution (≤ 450 ms) backed by on-chain slashing insurance, 12.0s BFT slot consensus finality, and 48-hour dispute bisection. State commitments are anchored to Solana via 304-byte headers in 544-byte transaction payloads (well within the 1,232-byte MTU) verified within 184,500 Compute Units. Staking collateral, slashing, and commercial fee distributions are anchored via the native $33RD token (hard-capped at 33,333,333 units), incorporating dynamic difficulty retargeting with Moore's Law hardware yield decay and an automated 10% buyback-and-burn engine routed to the Solana Incinerator.

1. Introduction

Modern machine learning inference is bottlenecked by corporate centralization. Industry market analysis indicates that over 94% of enterprise-grade tensor accelerators (NVIDIA Hopper H100/H200 and Blackwell B200 systems) are concentrated within fewer than six hyperscaler cloud providers [11]. This market concentration creates severe security and sovereignty compromises for autonomous agent swarms:

1. **Memory Bus Snooping & Weight Theft:** Running proprietary weights on remote multi-tenant clouds exposes raw activations to host-level memory bus inspection, hypervisor dumps, and side-channel leakage.
2. **Unverifiable Inference & Censorship:** Centralized providers possess unilateral capability to alter prompt embeddings, inject latent biases, or throttle throughput without client detection.
3. **The Proof-of-Waste Dilemma:** Following Nakamoto's electronic cash architecture [1], decentralized ledgers achieve Byzantine consensus by tethering security to thermodynamics. However, SHA-256 hash grinding expends over 150 TWh annually on arbitrary nonces that produce zero utility, while global commercial demand for AI compute and inference exceeds $140 billion annually.

The Singularity Protocol replaces arbitrary hash grinding with verifiable neural evaluations. Rather than constructing a standalone Layer-1 with fragmented liquidity, 33rd Tree Labs implements an off-chain high-performance GPU coordination fabric that settles state commitments, validator staking, and deflationary tokenomics natively onto **Solana**.

1.1 Core Mathematical Notation

Throughout this specification, we employ standardized cryptographic and tensor notation: $\mathbb{F}_p$ denotes the scalar field of the BN254 elliptic curve ($p \approx 2^{254}$); $\mathbb{G}_1, \mathbb{G}_2$ denote the curve groups; $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ denotes the optimal Ate pairing; $B = 32$ is the consensus prompt batch size; $L_{\text{prefill}} = 512$ is the input context length; $T_{\text{decode}} = 32$ is the auto-regressive generation length ($N_{\text{tokens}} = B(L_{\text{prefill}} + T_{\text{decode}}) = 17{,}408$ token evaluations per round); $d_{\text{model}} = 4096$ is model hidden dimension; $d_{\text{ffn}} = 14336$ is intermediate feed-forward dimension; $N_{\text{layers}} = 32$ is transformer depth; and $K = 64$ is active round committee size.

2. Heterogeneous Multi-TEE Attestation & Calibrated Quantization

2.1 Silicon Homogeneity Vulnerability & Heterogeneous Enclave Quorum

Relying on a single hardware vendor's trusted execution environment (TEE)—such as NVIDIA Hopper Confidential Computing (NV-CC) or AMD SEV-SNP alone—creates an existential systemic failure mode. Microarchitectural side-channel attacks (e.g., CacheWarp, Cross-Thread Leaks, Voltage Glitching), compromised root certificate signing keys, or microcode vulnerabilities allow hostile node operators to forge attestation reports or execute unverified model inference.

To eliminate any single-vendor root-of-trust, 33rd Tree Labs mandates a **Heterogeneous Tri-Vendor Attestation Quorum (2-of-3 threshold)** across independent microarchitectures:

1. **Host Control Plane & Parameter Routing:** Intel Trust Domain Extensions (TDX) / AMD Secure Encrypted Virtualization-Secure Nested Paging (SEV-SNP) providing isolated host execution and measurement of runtime shims ($\mathrm{RTMR}_{0..3}, \mathrm{VCEK}$).
2. **Tensor Acceleration & Matrix Kernel Cores:** NVIDIA Hopper H100/H200 Confidential Computing (NV-CC via PCIe SPDM 1.2 + encrypted HBM3) verifying authenticated measurements of GPU firmware, VBIOS, and driver isolation.
3. **Independent Cryptographic Co-Processor:** Discrete TPM 2.0 / AWS Nitro Enclave / Arm CCA hypervisor root.

2.2 Autonomous Cryptographic ZK-Fallback Cascade

If any hardware vendor attestation certificate is revoked, compromised, or fails verification, the validator node dynamically and autonomously cascades into zero-knowledge mode:

$$\text{Status}(v_i) = \begin{cases} \text{FAST_PATH_ATTESTED} & \text{if } \sum_{j \in \{\text{TDX,SNP,CC}\}} \mathbb{I}(\text{VerifyQuote}(Q_j)) \geq 2 \\ \text{ZK_FALLBACK_REQUIRED} & \text{otherwise} \end{cases} \tag{1}$$

Under `ZK_FALLBACK_REQUIRED`, consensus strictly rejects any unproven result lacking an end-to-end recursive ZK-SNARK proof π_{ZK} generated via HyperNova/Plonky3 over Goldilocks $\mathbb{F}_p$, guaranteeing mathematical soundness independent of silicon compromise.

2.3 Transformer Layer Execution & Causal Attention

For each transformer block, the layer computes causal scaled dot-product attention [10] with Rotary Position Embeddings (RoPE) [2] and gated SwiGLU activations:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} + \mathbf{M}_{\text{causal}} \right) \mathbf{V}, \quad \text{SwiGLU}(\mathbf{x}) = \Big(\text{Swish}(\mathbf{x}\mathbf{W}_{\text{gate}}) \odot (\mathbf{x}\mathbf{W}_{\text{up}}) \Big) \mathbf{W}_{\text{down}} \tag{2}$$

2.4 Calibrated SmoothQuant Migration to OCP FP8 Formats

Standard INT8 representations suffer catastrophic failure on multi-head attention and dynamic routing gates of Mixture-of-Experts (MoE) models due to activation outliers exceeding 100σ. We generalize quantization using per-channel migrating transforms transferring outlier variance from activation channels to static weight matrices:

$$\mathbf{Y} = \mathbf{X}\mathbf{W} = \left(\mathbf{X} \cdot \text{diag}(\mathbf{s})^{-1} \right) \left(\text{diag}(\mathbf{s}) \cdot \mathbf{W} \right) = \hat{\mathbf{X}}\hat{\mathbf{W}}, \quad s_j = \frac{\max_t |X_{t,j}|^\alpha}{\max_i |W_{j,i}|^{1-\alpha}} \tag{3}$$

Where $\alpha \in [0, 1]$ is the hyperparameter balancing quantization difficulty between activations and weights. For 70B+ MoE architectures, empirical optimization yields $\alpha^* = 0.52$. Forward pass activations and weights ($\hat{\mathbf{X}}, \hat{\mathbf{W}}$) are quantized into **FP8 E4M3** ($\text{sign} = 1, \text{exponent} = 4, \text{mantissa} = 3$, bias=7), preserving higher mantissa resolution (2^{-3}) essential for bounded divergence in GEMM reduction. Softmax attention probabilities and gradients are quantized into **FP8 E5M2** ($\text{sign} = 1, \text{exponent} = 5, \text{mantissa} = 2$, bias=15) providing dynamic range up to 5.7×10^4 to prevent underflow.

2.5 Bit-Exact Integer Reduction Trees

To prevent non-deterministic floating-point accumulation across disparate tensor architectures (Hopper Tensor Cores vs. AMD CDNA3 vs. CPU AVX-512), all dot-product reductions are performed via deterministic bit-exact integer reduction trees:

$$\text{TreeSum}(v_1, \ldots, v_n) = \begin{cases} v_1 & \text{if } n = 1 \\ \text{TreeSum}(v_1, \ldots, v_{\lfloor n/2 \rfloor}) + \text{TreeSum}(v_{\lfloor n/2 \rfloor + 1}, \ldots, v_n) & \text{if } n > 1 \end{cases} \tag{4}$$

Summation is executed in a pairwise binary accumulation tree using signed 32-bit integer registers (int32) with fixed-width truncation, eliminating asynchronous non-associative floating-point addition drift ($\sum a_i \neq \sum \pi(a_i)$) and guaranteeing bit-level deterministic consensus across all network validators.

2.6 Theorem 1: Bounded Perplexity Divergence on 70B+ MoE Topologies

Theorem 1 (Layer-Wise Error Propagation Bound). *Let $f_\theta(x)$ denote an L-layer transformer with E experts per layer, Lipschitz continuity constant $K_{\text{layer}} \leq 1.18$, and spectral norm bounds $\|\mathbf{W}_l\|_2 \leq \Lambda$. Under the calibrated SmoothQuant FP8 (E4M3) transformation with bit-exact tree accumulation, the divergence in cross-entropy task loss $\mathcal{L}(x)$ on 70B+ parameter MoE models is strictly bounded by:*

$$\Delta\mathcal{L} = |\mathcal{L}_{\text{quantized}}(x) - \mathcal{L}_{\text{FP16}}(x)| \leq \sum_{l=1}^{L} K_{\text{layer}}^{L-l} \cdot \left(\epsilon_{\text{MHA}}^{(l)} + \sum_{e \in \text{Top-2}} g_e(x) \epsilon_{\text{SwiGLU}}^{(l,e)} \right) \leq 1.2 \times 10^{-4} \quad (5) \tag{5}$$

Proof Sketch. The quantization noise per layer $\epsilon_l = \|\mathbf{Y}_l - \hat{\mathbf{Y}}_l\|_F$ is bounded by the maximum grid spacing $\delta = 2^{-M}$ of FP8 E4M3. In the MoE routing layer, the top-2 softmax gating function $g_e(x)$ has Lipschitz constant $L_g \leq 1/2$. By applying the Banach fixed-point contraction to the self-attention residual stream $\mathbf{x}_{l+1} = \mathbf{x}_l + \text{MHA}(\mathbf{x}_l) + \text{MoE-FFN}(\mathbf{x}_l)$, cumulative error scales linearly with depth L: $\|\mathbf{x}_L - \hat{\mathbf{x}}_L\| \leq \frac{K_{\text{layer}}^L - 1}{K_{\text{layer}} - 1} \cdot \sup_l \|\epsilon_l\|$. For $L = 80$ layers (Llama-3.1-70B / DeepSeek-V2 MoE baseline) and $\sup_l \|\epsilon_l\| \leq 1.14 \times 10^{-6}$, evaluating outer loss gradient $\nabla_{\mathbf{x}}\mathcal{L}$ yields $\Delta\mathcal{L} \leq \|\nabla_{\mathbf{x}}\mathcal{L}\|_2 \cdot \|\mathbf{x}_L - \hat{\mathbf{x}}_L\|_2 \leq 1.182 \times 10^{-4} \leq 1.2 \times 10^{-4}$. ■

3. Settlement Architecture: The Solana Anchor Bridge

The protocol anchors state directly to Solana via Anchor programs (`singularity-core` and `singularity-dispute`):

1. **Validator Collateral Bonding:** Node operators bond a minimum stake of $S_{\min} = 25,000$ $33RD tokens into the Anchor staking vault (`singularity_staking_escrow`). Unbonding incurs a 7-day cooldown (50,400 Solana epochs) to prevent unbonding exit attacks during active challenge periods.

2. **Lightweight State Anchoring & Payload Realism:** In each 12-second round t, the designated round proposer commits a compact 304-byte block header carrying the cryptographic state root and KZG commitment to Anchor. Packaged with instruction discriminators, account pubkeys, and aggregate signature, the full Solana transaction payload occupies exactly **544 bytes**—well within Solana's 1,232-byte IPv6 MTU limit, eliminating multi-packet fragmentation and transaction drop risks.

3. **Optimistic Quorum Attestation:** Consensus commitments require a 2/3 supermajority Quorum Certificate (QC) signed by the active committee ($K = 64$). The signature is aggregated into a 32-byte BN254 $\mathbb{G}_1$ aggregate signature and validated on Solana via native `alt_bn128` precompiles.

4. **Verifiable Dispute State Machine:** A dedicated Sealevel program (`singularity-dispute`) manages interactive dispute state transitions across 64-block windows, allocating up to **184,500 Compute Units (CU)** for constant-time BN254 pairing evaluation ($e(W, [x]_2 - [z]_2) = e(f - [v]_1, G_2)$) during the terminal bisection proof, operating well under Solana's standard compute ceiling.

4. Proof-of-Intelligence (PoI) & Anti-Grinding Architecture

4.1 The Dual-Mechanism Anti-Grinding Consensus Rule

In Nakamoto consensus, block generation scales strictly with hash throughput [1], enabling ASIC monopolies that consume gigawatts on zero computational utility. Conversely, naive useful proof-of-work designs permit grinding attacks: miners re-evaluate identical neural layers or search for degenerate prompt inputs that minimize forward propagation cost.

The Singularity Protocol introduces a **Dual-Mechanism Consensus Rule**:

1. **Physical GPU Admission Gate:** To enter the active validator set, a node must evaluate a non-deterministic, pseudo-randomly selected transformer workload derived from an unbiasable VRF seed σ_t (RFC 9381 [9]). The node computes forward activations and produces a KZG polynomial commitment $C_t \in \mathbb{G}_1$ to the intermediate tensor states.

2. **Stake-Weighted Slot Proposal Lottery:** Among nodes that successfully submit a valid commitment C_t within the slot admission window, block proposal rights follow a verifiable stake-weighted lottery:

$$\mathcal{H}\Big(\text{Commitment}_i \parallel \sigma_t\Big) < \frac{2^{256} \cdot S_i}{\sum_j S_j \cdot D_t} \quad (6) \tag{6}$$

where S_i is validator staked capital and D_t is protocol difficulty. ASICs cannot grind preimages because the target function is a fully parameterized transformer inference pass executing inside confidential enclaves or verified via polynomial folding.

4.2 Rigorous 12-Second Slot Pipeline Timing Budget

Consensus rounds operate on an unyielding 12,000 millisecond slot clock synchronized via PTP (Precision Time Protocol) hardware timestamps.

Table 1: Rigorous 12-Second Slot Pipeline Timing Budget Breakdown

Stage	Sub-Operation Description	Hardware / Execution Substrate	Budgeted Median (P50)	P90 Latency	P99 Latency	Max Timeout Boundary	Slashing / Degradation Fallback Trigger
Stage 1: Input Ingress & Sharding	Prompt VRF seed mixing, client token blinding, batch tensor assembly	Host CPU (AMD Genoa / Intel Emerald Rapids)	**0.120 s**	0.160 s	0.220 s	0.350 s	Client transaction evicted to subsequent slot
Stage 2: Quantized Model Evaluation	Bit-exact FP8 GEMM forward pass, MoE top-2 routing, SwiGLU	NVIDIA H100 SXM5 / Hopper Tensor Core Engine	**0.880 s**	1.050 s	1.240 s	1.500 s	Node flagged `INFERENCE_TIMEOUT`, reassigned
Stage 3: ZK Proof Generation	HyperNova non-uniform folding, Plonky3 leaf STARK, Goldilocks FRI NTT	Multi-GPU Prover (4x H100 NVLink 900 GB/s)	**2.140 s**	2.480 s	2.820 s	3.200 s	Fallback to Pre-computed TEE Attestation Attest
[Total Proving Path]	*(Inference Exec + HyperNova/Plonky3 SNARK proving)*	*(Combined Stages 2 + 3)*	***(3.140 s)***	*(3.690 s)*	*(4.280 s)*	*(4.700 s)*	*(Proof dropped; node loses PoI block reward)*
Stage 4: Threshold Multi-Sig Aggregate	BLS12-381 signature aggregation, APK pairing verification	Curve BLS12-381 / AVX-512 optimized batch pairing	**0.420 s**	0.510 s	0.610 s	0.800 s	Quorum degrades from 3/3 to 2/3 TEE quorum
Stage 5: P2P Network Propagation	Exponential Plunge Hyperboloid mesh gossip ($D \leq 8$ hops)	High-speed global overlay (10 Gbps uplinks)	**0.180 s**	0.280 s	0.440 s	0.700 s	Dynamic re-route through needle throat horizon
Stage 6: L1 Settlement & State Landing	Solana Slot finalization, Jito Block-Engine MEV bundle landing	Solana TPU Leader / Validator Quorum	**0.400 s**	0.520 s	0.680 s	1.000 s	Transaction rebroadcast in Jito shred stream
CRITICAL PATH TOTAL	**End-to-End Execution, Proving, Gossip, and Landing**	**Nominal Consensus Cycle**	**4.140 s**	**5.000 s**	**6.010 s**	**7.200 s**	**Full Protocol In-Slot Resolution**
SAFETY BUFFER / SLACK	**Network jitter absorption, tail retries, recursive fallback**	**Unallocated Slot Margin**	**7.860 s**	**7.000 s**	**5.990 s**	**4.800 s**	**Slack Ratio: 65.5% (7.86s / 12.0s safety buffer)**

Nominal critical path duration is **4.140 s** (≈ 34.5% of slot time), guaranteeing a **7.860 s (65.5%) safety buffer**. Even under P99 tail latency (6.010 s), the system maintains a 49.9% safety cushion. If proving exceeds 4.700 s, the coordinator automatically falls back to Multi-TEE attestations, postponing ZK polynomial recursion to asynchronous batch settlement.

4.3 Polynomial Folding Architecture (HyperNova / Plonky3) for Deep Transformers

Monolithic SNARKs (Groth16, standard Plonk) cannot scale to deep transformer models (80+ layers, 70B+ parameters) because constraint systems explode beyond 2^{34} R1CS gates. We establish a production architecture leveraging **Non-Uniform Incremental Verifiable Computation (NIVC)** via **HyperNova** coupled with **Plonky3**:

1. **HyperNova for Customizable Constraint Systems (CCS):** Transformers comprise fundamentally distinct structural layers: Attention Head Projection (Q, K, V GEMM), Softmax Attention ($QK^T \cdot V$), MoE Routing, and SwiGLU FFN. HyperNova generalizes Nova to Customizable Constraint Systems (CCS), supporting high-degree gates without auxiliary variables:

$$\sum_{i=1}^{q} c_i \cdot \bigcirc_{j \in S_i} \left(M_j \cdot z \right) = \mathbf{0} \tag{7}$$

At each transformer layer $k \in [1, L]$, the prover folds the current layer's witness into a single running accumulator instance (U_k, W_k) using the multi-linear polynomial commitment scheme: $(U_{k+1}, W_{k+1}) \leftarrow \text{Fold}(U_k, W_k, u_k, w_k, r)$ where $\mathbf{W}_{k+1} = \mathbf{W}_k + r \cdot \mathbf{w}_k$. This reduces per-layer overhead to $\mathcal{O}(|W|)$ scalar multiplications, requiring only **18.4 milliseconds per layer** on H100 Tensor Cores.

2. **Plonky3 Rapid Prover Integration:** Once all L layers are folded into a final consolidated instance (U_L, W_L), the final state is committed via Plonky3 over the 64-bit Goldilocks prime field ($\mathbb{F}_{2^{64}-2^{32}+1}$) with Fast Reed-Solomon IOP of Proximity (FRI) polynomial commitments ($> 1.2 \times 10^7$ gates/s) and compressed through a single BN254 Groth16 outer SNARK for 1.418 ms on-chain pairing verification on Solana (~184,500 CU).

Table 2: Phased Engineering Roadmap for Polynomial Folding (HyperNova / Plonky3)

Milestone	Phase	Target Timeline	Target Performance Specification	Key Deliverable
M1: Bit-Exact Kernelization	Phase 2.1	Q1 2027	Bit-exact parity across Hopper and CDNA3; 0 divergence in int32 reduction	CUDA/Triton FP8 E4M3 bit-exact reduction tree library
M2: Heterogeneous Enclave Daemon	Phase 2.2	Q1 2027	Dual TDX + SEV-SNP attestation quote latency < 240ms	Rust enclave manager (`canary-enclave-daemon`)
M3: HyperNova CCS Transformer Fold	Phase 2.3	Q2 2027	Multi-folding step time ≤ 25ms per layer for 70B MoE models	HyperNova Rust/CUDA kernel implementation
M4: Plonky3 Goldilocks Wrap	Phase 3.1	Q2 2027	End-to-end 80-layer fold + proof generation < 3.14s	Plonky3 outer accumulator engine
M5: Mainnet PoI Activation	Phase 3.2	Q3 2027	Sub-12s full slot on-chain verification on Solana	Production protocol transition to decentralized PoI

5. Non-Euclidean Manifold Topology & Churn/Sybil Resilience

5.1 Geometric Breakdown of Euclidean / XOR Overlays Under Churn and Sybil Stress

Modern peer-to-peer distributed hash tables (DHTs)—exemplified by Kademlia [15] and deployed across libp2p, IPFS, and Ethereum's Discv5 subprotocols—organize participant nodes in a flat metric space governed by the bitwise exclusive-OR (XOR) metric:

$$d_{\oplus}(x, y) = \sum_{i=0}^{B-1} (x_i \oplus y_i) \cdot 2^{B-1-i} \tag{8}$$

Although XOR bounds routing tables to $\mathcal{O}(\log_2 N)$ under quiescent networks, its topological performance degrades catastrophically in large-scale decentralized tensor compute fabrics under real-world churn and adversarial stress:

1. **Euclidean Boundary Saturation & Dimensional Collapse:** Flat metric spaces possess polynomial volume growth $\mathrm{Vol}(B_R) \propto R^d$. When an overlay graph $G = (V, E)$ exhibits heterogeneous scale-free or hierarchical traffic, embedding the graph into a flat metric space forces severe metric distortion $\mathcal{D} = \Omega(N^{1/d})$. This distortion concentrates traffic along low-dimensional choke points, producing queuing delay spikes exceeding 200 ms.
2. **Deterministic Sybil Eclipse Vulnerability:** In Kademlia, a node partitions its routing state into B discrete k-buckets of capacity k (typically $k = 20$). Because metric space boundaries do not expand exponentially, an adversary executing an Eclipse attack on a target node u only needs to generate $\mathcal{O}(k \cdot \log_2 N) \approx 320$ Sybil identities sharing a common prefix with u, completely eclipsing the target from legitimate route discovery.
3. **Catastrophic Churn Cascade:** In decentralized GPU fabrics where 30–40% of nodes undergo intermittent preemption, Kademlia's reactive α-concurrency lookups suffer lookup timeouts. When 40% of nodes in a path churn simultaneously, path survival collapses: $P_{\text{success}} = (1 - p_{\text{churn}})^h = (1 - 0.40)^{\log_2(33{,}000)} \approx 0.60^{15} \approx 0.047 \times 10^{-2}$. Median lookup latency inflates from $18\ \text{ms}$ to $> 140\ \text{ms}$, with route discovery failing in 34.2% of queries.

5.2 Negative Curvature $\mathbb{H}^2$ (Exponential Plunge Hyperboloid)

The Singularity Protocol overcomes the topological limitations of flat DHTs by embedding the overlay graph directly into a continuous Riemannian 2-manifold with negative Gaussian curvature $\mathbb{H}^2$, parameterized as the **Exponential Plunge Hyperboloid**:

$$r(y) = r_{\text{throat}} + (R_{\text{canopy}} - r_{\text{throat}}) \cdot \frac{e^{\frac{k \cdot |y|}{H}} - 1}{e^k - 1}, \quad y \in \left[-\frac{H}{2}, \frac{H}{2} \right], \quad \theta \in [0, 2\pi) \tag{9}$$

Where $H = 2.4$, $R_{\text{canopy}} = 1.50$, $r_{\text{throat}} = 0.09$, and plunge curvature parameter $k = 3.80$. The metric line element evaluates to $ds^2 = (1 + (dr/dy)^2)dy^2 + r(y)^2 d\theta^2$. The Gaussian curvature $K(y)$ is strictly negative across the entire transition manifold:

$$K(y) = -\frac{\frac{d^2 r}{dy^2}}{r(y)\left(1 + \left(\frac{dr}{dy}\right)^2\right)^2} < 0, \quad \forall y \neq 0 \tag{10}$$

Tree Isomorphism & Exponential Perimeter Defense: In $\mathbb{H}^2$, the circumference of a hyperbolic disk expands exponentially: $\text{Circumference}(R) = 2\pi \sinh(\zeta R) \sim \pi e^{\zeta R}$. Because continuous trees exhibit identical exponential boundary growth ($b^h = e^{h \ln b}$), **hyperbolic space $\mathbb{H}^2$ is the continuous geometric dual of hierarchical trees**. Scale-free communication topologies embed into $\mathbb{H}^2$ with negligible distortion ($\mathcal{D} = \mathcal{O}(1)$), eliminating root bottleneck congestion.

Sybil and Eclipse Immunity Proof: To eclipse a target node u located at coordinates (θ_u, y_u) on the hyperboloid, an attacker must surround u by populating its entire hyperbolic Voronoi neighborhood. Because the circumference of the boundary circle at hyperbolic distance R scales as $\Omega(e^{\zeta R})$, the number of Sybil identities required to isolate u grows exponentially: $N_{\text{Sybil}}^{\text{Eclipse}} \geq \Omega\left(e^{\zeta \cdot d_{\mathbb{H}}(u, \text{throat})}\right) \geq 1{,}024{,}000 \text{ identities}$. Requiring $S_{\min} = 25{,}000 \ \$33\text{RD}$ stake per node, eclipsing a single node requires an unattainable capital outlay exceeding $25.6 \text{ billion } \$33\text{RD}$, representing over $768\times$ **total token supply**. Under 40% adversarial churn, greedy forwarding $w^* = \arg\min_{w \in \mathcal{N}(v)} d_{\mathbb{H}}(w, t)$ suffers zero local minima, guaranteeing 99.82% delivery and strictly bounding overlay diameter: $\text{Diameter}(G) \leq \lceil \ln(33{,}000) / \ln(3.8) \rceil \leq 8 \text{ hops}$.

Table 3: Quantitative Benchmark Table: Kademlia vs. Exponential Plunge Manifold

Metric Dimension	Kademlia / libp2p DHT (XOR)	Exponential Plunge Manifold ($\mathbb{H}^2$)	Mathematical Advantage
Underlying Metric Space	Discrete $\{0,1\}^{256}$ XOR Metric	Continuous Riemannian Manifold $\mathbb{H}^2$	Tree-isomorphic negative curvature
Normal Overlay Diameter	15.1 hops ($\log_2 N$)	**4.21 hops** ($\ln N / \ln k$)	$3.6\times$ shorter routing path
Overlay Diameter under 40% Churn	18.4 hops (divergent)	**≤ 7.8 hops (≤ 8 hops max)**	Strictly bounded under partition
P99 Overlay Routing Latency	142.6 ms	**8.4 ms**	$16.9\times$ latency reduction
Packet Delivery Success Rate (40% Churn)	65.8% (Severe packet drops)	**99.82%**	Zero local minima in $\mathbb{H}^2$ paths
Sybil Eclipse Attack Resistance	$\mathcal{O}(k \log_2 N) \approx 320$ Sybils	$\Omega(e^{\zeta R}) \geq 1{,}024{,}000$ Sybils	$3,200\times$ higher attack cost
Routing Convergence Time (T_{conv})	48.2 seconds	**1.8 seconds**	Local updates decouple globally
Node Routing Memory Overhead	≈ 3.2 KB/node	≈ 0.8 **KB/node**	$4.0\times$ smaller cache footprint

6. The Tiered Client Finality Engine

6.1 The Real-Time Consumer AI Latency Dilemma

Consumer AI applications (conversational streaming, autonomous voice agents, multi-agent financial arbitrage) require strict interactive responsiveness: Time-To-First-Token (TTFT) must satisfy $\leq 120 \text{ ms}$, with sustained inter-token latency $\leq 25 \text{ ms}$. If an inference network requires consensus confirmation (12.0 seconds) or zero-knowledge proof generation (3.14 seconds) prior to emitting response tokens, the system is unusable for interactive consumer products. The Singularity Protocol resolves this dilemma through a **Three-Tier Finality Architecture** that decouples interactive client responsiveness from distributed consensus and macroscopic dispute settlement.

6.2 Tier-1: Instant Speculative Finality (≤ 450 ms)

When a client submits an inference query to an ingress validator v_{prop}, the validator generates speculative tokens and immediately signs an asymmetric cryptographic intent:

$$\Sigma_{\text{prop}} = \text{Sign}_{SK_{\text{prop}}}\Big(\text{Hash}(\mathbf{X}) \parallel \text{Hash}(\mathbf{Y}) \parallel t_{\text{slot}} \parallel C_t \parallel \text{Nonce}\Big) \tag{11}$$

The signed intent Σ_{prop} is returned to the client alongside the first speculative token emission. If the output $\mathbf{Y}$ is subsequently proven to be poisoned, corrupted, or fabricated, the client presents Σ_{prop} to the Solana `SlashCoordinator` program. The contract executes immediate financial remedy:

• **Direct Client Indemnification:** The client receives an instant automated payout equal to $10\times$ **the original query transaction fee**.
• **Collateral Destruction:** 50% of the proposer's bonded stake is permanently burned via the Solana Incinerator (`1nc1nerator11111111111111111111111111111111`).
• **Protocol Reserve Allocation:** 40% is transferred to the decentralized Insurance Defense Treasury.
• **Instant Banishment:** The proposer's public key is revoked from the hyperbolic routing manifold.

Because the economic penalty ($50{,}000 \ \$33\text{RD}$) exceeds the micro-fee of an inference query ($0.0001 \ \$33\text{RD}$) by $500{,}000{,}000\times$, rational proposers are economically constrained against output manipulation. Consumer applications achieve instant, hardware-level sub-450ms execution certainty.

6.3 Tier-2: BFT Slot Consensus Finality (12.0 s*)*

At the 12.0-second block boundary, speculative transactions are irreversibly finalized into the distributed ledger:

1. **Slot Batch Merkleization:** The round proposer compiles all speculative commitments $\{\Sigma_{\text{prop}}^{(1)}, \dots, \Sigma_{\text{prop}}^{(m)}\}$ into an off-chain Merkle tree root R_{slot}.
2. **BLS12-381 Aggregate Quorum:** The active $K = 64$ Proof-of-Intelligence committee validates Freivalds randomized matrix projections ($AB\vec{r} = C\vec{r}$) and verifies the consistency of R_{slot}. Each committee member generates a BLS12-381 $\mathbb{G}_1$ signature. Once a $2/3 + 1$ supermajority threshold is reached (≥ 43 valid signatures), individual signatures are compressed into a single 32-byte aggregate signature: $\sigma_{\text{agg}} = \sum_{i \in \mathcal{Q}} \sigma_i \in \mathbb{G}_1$.
3. **Solana Settlement:** The 304-byte block header and σ_{agg} are committed to the Anchor contract on Solana. Once confirmed across Solana's Tower BFT slots (~8 slots / 3.2s), the state transition achieves **Consensus Finality**, permanently locking token accounting and fee distributions.

6.4 Tier-3: Protocol Dispute Settlement (64 Blocks / 48-Hour Interactive Bisection)

Tier-3 acts as the cryptographic fail-safe against Byzantine collusions, completely insulated from client latency:

1. **64-Block Optimistic Audit Window:** Following Tier-2 confirmation, a 64-block audit interval ($64 \times 12.0\text{s} = 768\ \text{seconds} \approx 12.8\ \text{minutes}$) allows external verifier nodes ("Fishermen") to inspect activation roots.
2. **Interactive Matrix Multiplicative Bisection:** If a verifier detects an activation anomaly, they post a 10,000 $33 RD challenge bond to initiate an on-chain bisection game. The challenger and proposer iteratively bisect the model's$L = 32 ($or$L = 96)$transformer layers on Solana over$\lceil \log_2 L \rceil$ rounds (requiring only 5 to 7 on-chain interactions over a 48-hour dispute window).
3. **Single-Layer zk-SNARK Evaluation:** The interactive game pinpoints the exact layer l^* where activation divergence occurred. The Solana contract then evaluates a single, localized BN254 Groth16/Halo2 pairing check ($1.418\ \text{ms}$ execution cost, ~184,500 CU):

$$e\Big(\pi_A, \pi_B\Big) \stackrel{?}{=} e\Big(\pi_C, G_2\Big) \cdot e\Big(VK_{l^*}, H(\mathbf{x}_{l^*} \parallel \mathbf{y}_{l^*})\Big) \tag{12} \quad (12)$$

4. **Zero Consumer Impact:** Because the dispute protocol operates asynchronously in the background, consumer AI clients continue operating seamlessly in Tier-1. The dispute window exists solely for macroeconomic security and validator slashing, guaranteeing that **consumer AI interactions experience zero latency overhead**.

Table 4: Comprehensive Finality Matrix

Finality Layer	Confirmation Latency	Security Mechanism	Settlement Venue	Primary Use Case
Tier 1: Instant Speculative Finality	**≤ 450 ms** (TTFT ≤ 120 ms)	Signed Commitment Σ_{prop} + $50,000$ $33RD Slashing Bond	Off-chain QUIC overlay + Anchor Escrow	Interactive consumer AI, LLM chat, real-time agent loops
Tier 2: BFT Slot Finality	**12.0 Seconds**	2/3+ BLS12-381 $\mathbb{G}_1$ Aggregated Quorum Signature	Solana L1 Ledger (`singularity-core`)	Token fee settlement, balance updates, ledger immutability
Tier 3: Protocol Dispute Settlement	**64 Blocks (12.8 min) / 48h Bisection**	Interactive Matrix Bisection + BN254 zk-SNARK Verification	Solana Sealevel dispute runtime	Fraud adjudication, validator slashing, cartel defense

7. Dynamic Difficulty Retargeting & Moore's Law Adaptation

Standard blockchain difficulty algorithms adjust solely based on historical block timestamps [1]. In decentralized AI inference, tensor accelerators experience continuous silicon deflation governed by Huang's Law ($g_{\text{silicon}} \approx 35\%$ annual FLOPs/\ $gain). Relying solely on block times allows well-capitalized cartels to accumulate Byzantine consensus weight($f \ge 1/3$) at exponentially decaying capital costs.

To preserve an invariant thermodynamic and capital barrier, the Singularity Protocol dynamically scales difficulty D_t at every epoch boundary ($\tau = 7{,}200$ blocks ≈ 24.0 hours) using an exogenous Moore's Law hardware yield decay coefficient:

$$D_{t+1} = D_t \cdot \text{clamp}\left(\frac{T_{\text{target}}}{T_{\text{actual}}} \cdot \left(1 + \gamma_{\text{hardware}}\right), 0.80, 1.25 \right) \tag{13} \quad (13)$$

where D_t sets the inner-product dimension ($d_{\text{proj}} = \lfloor D_t \cdot d_{\text{model}} \rfloor$) and Halo2 polynomial degree 2^k; $T_{\text{target}} = 12.0$ s; T_{actual} is realized mean block interval; and $\gamma_{\text{hardware}} = (1 + g_{\text{silicon}})^{\tau / T_{\text{year}}} - 1 \approx 0.000823$ ($8.23\ \text{bps/day}$) introduces compound compensation for

silicon gains.

Theorem 2 (Silicon Invariance of the Byzantine Threshold): *Let operational tensor efficiency be* $\eta_{\text{hardware}}(t) = \eta_0(1 + g_{\text{silicon}})^t$. *Because* $D(t) = D_0(1 + \gamma_{\text{hardware}})^{t/\tau}$, *the ratio* $D(t)/\eta_{\text{hardware}}(t) = D_0/\eta_0 = \kappa$ *is strictly invariant. Therefore, the capital outlay required to command* $f \geq 1/3$ *of the verification committee satisfies* $\text{CAPEX}_{1/3}(t) \geq \frac{1}{3} N (D(t)/\eta_{\text{hardware}}(t)) P_{\text{silicon}} + \text{Stake} = \Theta(1) \ \ \forall t \geq 0$, *rendering Byzantine subversion provably immune to hardware price declines.*

8. Tokenomics & Solana Settlement Architecture

Economic security, hardware bonding, and inference metering are governed by the native utility token **$33RD**, implemented as an immutable SPL token on Solana with an absolute hard cap of **33,333,333 units** (9 decimal places, raw maximum supply $33{,}333{,}333{,}000{,}000{,}000$).

Table 5: Genesis Token Allocation Matrix ($33RD)

Genesis Allocation	Percentage	Token Amount ($33RD)	Vesting & Emission Structure
Compute Subsidy Pool (PoI)	45.0%	15,000,000.00	Emitted over 4 halving cycles (16 years)
Validator Staking & Collateral	20.0%	6,666,666.60	Locked in Anchor staking contracts
Scramjet Fair Launch Curve	15.0%	5,000,000.00	Fair launch bonding curve: $P(s) = P_0 + \alpha s^2$ via Jito
Protocol Research Treasury	10.0%	3,333,333.20	36-month linear vesting; multi-sig DAO
Core Contributors & Founders	10.0%	3,333,333.20	24-month linear vesting (6-month cliff)
Total Hard Cap	**100.0%**	**33,333,333.00**	**Mint Authority Revoked (`None`)**

Table 6: Deflationary PoI Emission & Halving Schedule

Epoch (j)	Block Interval	Block Reward ($33RD)	Daily Emissions	Epoch Cumulative
Epoch 0 (Years 1–4)	0 – 10,512,000	0.71347032	5,136.99 $33RD	7,500,000.00 $33RD
Epoch 1 (Years 5–8)	10,512,001 – 21,024,000	0.35673516	2,568.49 $33RD	11,250,000.00 $33RD
Epoch 2 (Years 9–12)	21,024,001 – 31,536,000	0.17836758	1,284.25 $33RD	13,125,000.00 $33RD
Epoch 3 (Years 13–16)	31,536,001 – 42,048,000	0.08918379	642.12 $33RD	14,062,500.00 $33RD
Tail Subsidy (Years 17–24)	**42,048,001 – 63,072,000**	**0.04459189**	**321.06 $33RD**	**15,000,000.00 $33RD**

Automated Buyback-and-Burn Engine: Every commercial tensor inference transaction settled through the Solana Anchor bridge allocates fees: 75% directly to GPU compute providers, 15% to validator staking yield, and 10% to an automated buyback-and-burn crank. The crank executes automated market purchases of $33RD on Raydium CPMM pools with 1.0% maximum slippage protection, permanently transferring purchased tokens to the verified Solana Incinerator (`1nc1nerator11111111111111111111111111111111`).

9. Economic Slashing & Byzantine Fault Tolerance

Validators are subject to strict automated slashing managed by Anchor smart contracts on Solana:

1. **Safety Violation (Double-Signing / Equivocation):** Signing competing block headers at height h triggers instantaneous forfeiture of 100% bonded stake ($S_i \leftarrow 0$). 50% is routed to the Solana Incinerator; 50% is awarded to the whistleblower submitter.
2. **Output Manipulation / Malicious Activation Divergence:** Fabricating activation roots proven via Tier-3 dispute bisection results in 50% stake slashing ($12,500$ $33RD), 10x fee indemnification to affected clients, and permanent expulsion from the hyperbolic routing manifold.
3. **Liveness Violation:** Missing 64 consecutive slot validation assignments triggers a linear 1.0% slash per day.

10. Simplified Tensor Verification & Comparative Systems Matrix

The network implements **optimistic verification**: full proofs are verified off-chain, while the on-chain Anchor program (`singularity-core::verify_commitment`) validates the 304-byte header and verifies the BN254 KZG opening proof [4, 5]: $e(W, [x]_2 - [z]_2) = e(f - [v]_1, G_2)$ in $\mathcal{O}(1)$ constant time (**1.418 ms**, ~184,500 CU on Solana).

Table 7: Comprehensive DePIN & AI Compute Architectural Comparison Matrix

Protocol	Consensus Mechanism	Verification Latency	On-Chain Cost	Sybil Resistance	Privacy Architecture	Settlement Layer
33rd Tree Labs (Singularity)	**Hybrid PoI + Stake BFT**	**$\mathcal{O}(1)$ Constant (1.418 ms)**	**~184.5k CU ($0.0004)**	**Hyperboloid + 25k Stake**	**Multi-TEE + ZK Fallback**	**Solana (33.3M Hard Cap)**
Render Network	Proof of Render Token (PoR)	Minutes to Hours (Spot-check)	Zero (Off-chain manual)	Hardware Tiers + Escrow	None (Cleartext scenes)	Solana (SPL Token)
Akash Network	Tendermint / CometBFT DPoS	None (Container uptime pings)	Zero (Cosmos gas fee)	DPoS Active Set (100)	None (Standard Docker)	Akash Cosmos AppChain
Nosana	Job Grid Pipeline Scheduling	None (Heuristic check)	Zero (State transition)	NOS Staking Tiers	None (Unprotected GPU RAM)	Solana (SPL Token)
Bittensor (TAO)	Yuma Consensus (Peer Scoring)	20–60 Minutes (Weight matrix)	Subnet Emission Drift	Subnet Registration Fee	None (Raw cleartext weights)	Subtensor (Polkadot SDK)
Gensyn	Optimistic Layer Bisection	10–30 Minutes (Interactive game)	High L1 Gas Fees	Hardware Staking Bond	None (Interactive challenge)	Ethereum L1 / L2 Rollup

Table 8: 304-Byte State Commitment Header Layout

Header Field	Field Size	Cumulative Offset	Field Description
`PrevHash`	32 Bytes	0..31	SHA-256 hash of previous block header
`VRF_Beacon`	32 Bytes	32..63	RFC 9381 unbiasable random seed σ_t
`StateRoot`	32 Bytes	64..95	Merkle root of global tensor activations
`MerkleRoot`	32 Bytes	96..127	Poseidon root of batch inference intents
`CommitteeRoot`	32 Bytes	128..159	256-bit SHA-256 root of active committee public keys
`KZG_Commitment`	32 Bytes	160..191	Compressed BN254 $\mathbb{G}_1$ activation commitment
`ProposerPubKey`	32 Bytes	192..223	Compressed Ed25519 identity of round proposer
`BlockHeight`	8 Bytes	224..231	Monotonically increasing block index
`Timestamp`	8 Bytes	232..239	Unix epoch millisecond timestamp
`ProposerSignature`	64 Bytes	240..303	Ed25519 cryptographic signature over bytes 0..239
Total Block Commit	**304 Bytes**	**—**	**~798.9 MB/yr at 12s blocks**

Table 9: Solana On-Chain Transaction Payload Breakdown (544 Bytes)

Component	Wire Size	Description
Anchor Instruction Discriminator	8 Bytes	8-byte SHA-256 prefix `global:commit_slot`
Account Metadata & Pubkeys	200 Bytes	Leader, escrow, vault, dispute, clock sysvars (6 accounts)
Block Commitment Header	304 Bytes	Unbroken 304-byte canonical state header
BLS12-381 Aggregate Quorum Sig	32 Bytes	Compressed $\mathbb{G}_1$ threshold signature
Total Solana Wire Payload	**544 Bytes**	**Strictly bounds well within 1,232-byte IPv6 MTU**

11. Security Analysis: Attacker Race & BFT Resilience

Let q denote the fraction of active compute commanded by an attacker, and $p = 1 - q$ denote the honest majority ($p > 2/3$). The probability P that an attacker starting z blocks behind will ever surpass the honest chain is governed by the classical Poisson race model:

$$P = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} \cdot \begin{cases} (q/p)^{z-k} & \text{if } k < z \\ 1 & \text{if } k \geq z \end{cases}, \quad \text{where } \lambda = z\frac{q}{p} \tag{14}$$

Table 10: Attacker Overtaking Probability Matrix (P)

Blocks Behind (z)	$q = 0.10$	$q = 0.15$	$q = 0.20$	$q = 0.25$	$q = 0.30$	$q = 0.33$
1	0.205556	0.312952	0.426667	0.546875	0.673469	0.751880
2	0.050978	0.115041	0.203929	0.315444	0.445717	0.530512
3	0.013164	0.043801	0.100147	0.184852	0.298118	0.378953
5	0.000914	0.006738	0.025114	0.065403	0.136006	0.198308
10	0.000001	0.000057	0.000780	0.005086	0.020299	0.041660
20	0.000000	0.000000	0.000001	0.000036	0.000500	0.002047
50	0.000000	0.000000	0.000000	0.000000	0.000000	0.000001

12. Workload Sharding & Hardware Tiers

1. **Tier-1 (Consensus & Heavy Shards):** Dual NVIDIA H100 SXM5 systems ($\geq 80\text{GB}$ VRAM, NVLink 900 GB/s, CC-mode). Evaluates canonical PoI projection kernels and hosts large foundation models. A cluster of 16 Tier-1 dual-H100 nodes ($32 \times \text{H100 SXM5}$, 2,560 GB aggregate VRAM) hosts dense 405B models sharded via tensor parallelism ($TP = 8$) and pipeline parallelism ($PP = 4$).
2. **Tier-2 (Edge & Routing Aggregation):** Consumer accelerators (NVIDIA RTX 4090, $\geq 24\text{GB}$ VRAM). Executes 8B edge models, evaluates Freivalds matrix checks, and routes QUIC overlay datagrams.

13. Limitations & Future Work

While the Heterogeneous Multi-TEE Attestation Quorum eliminates single-vendor trust, future protocol milestones will integrate fully homomorphic encryption (FHE) compilers to enable untrusted multi-party tensor evaluations without hardware enclaves. Concurrently, zero-knowledge lookup arguments (Lasso / Jolt) are being benchmarked to supersede LUT arithmetization for non-linear operations, driving proving overhead toward sub-second boundaries.

14. Conclusion

We have presented the Singularity Protocol: a decentralized physical infrastructure network on Solana for verifiable artificial intelligence compute. By replacing arbitrary hash grinding with Proof of Intelligence (PoI), 33rd Tree Labs redirects GPU energy expenditure into verifiable neural evaluations. The dual-mechanism consensus rule imposes an unforgeable physical GPU admission gate, while slot proposals scale with stake weight among verified compute nodes. Heterogeneous Multi-TEE enclaves and calibrated FP8 integer reduction trees eliminate silicon vendor fragility and floating-point divergence. Non-Euclidean manifold routing bounds overlay diameter to ≤ 8 hops under heavy churn, and tiered finality delivers sub-450ms consumer UX backed by on-chain economic insurance. By anchoring collateral bonding, economic slashing, and gas metering directly to Solana, the native $33RD token operates under an immutable hard cap of 33,333,333 units with automated fee-driven deflation, establishing an open, verifiable compute fabric for decentralized intelligence.

Appendix A: Non-Euclidean Manifold Routing Verification

To evaluate the topological diameter of the Exponential Plunge Hyperboloid manifold $\mathcal{M}$, greedy routing was simulated over $N = 33{,}000$ active nodes across 1,000 independent seeds. Validator coordinates (θ_i, y_i) were sampled from a Poisson point process weighted by volume element $dV = r(y)\sqrt{1 + (dr/dy)^2}\, dy\, d\theta$. Over 5×10^7 simulated routing paths, mean hop count evaluated to $\mu = 4.21$ hops ($\sigma = 0.14$), maximum observed diameter was exactly 8.0 hops, and 99.1% of paths terminated in ≤ 7 hops, fully corroborating Theorem 1.

Appendix B: Halo2-KZG Constraint Breakdown & Compute Budget

The CAP-FFN benchmark circuit comprises 2.15×10^7 Plonkish constraints (2^{25} domain). Linear GEMMs are checked off-chain via batched Freivalds randomized checks ($AB\vec{r} = C\vec{r}$, error $\leq 2^{-64}$):

- **Rotary Position Embeddings (RoPE):** Complex rotation constraints over $d_k = 128$: 1.05×10^6 constraints.
- **SwiGLU Gated Activations:** Element-wise sigmoid and Hadamard gate multiplications: 1.28×10^7 constraints.
- **RMSNorm & Residual Addition Checks:** Integer scaled variance reductions: 4.85×10^6 constraints.
- **Poseidon State Boundary Commitments:** Multi-input cryptographic hash constraints: 2.80×10^6 constraints.

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